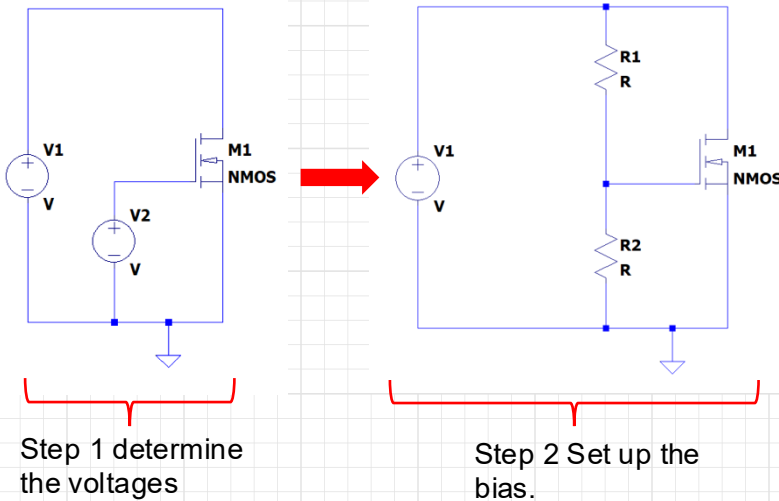


Lab 2: Amplifiers

Task 7: Biasing a MOSFET transistor



In this first circuit we aim to bias an nMOS transistor such that you attain an output current of $40\mu\text{A}$.

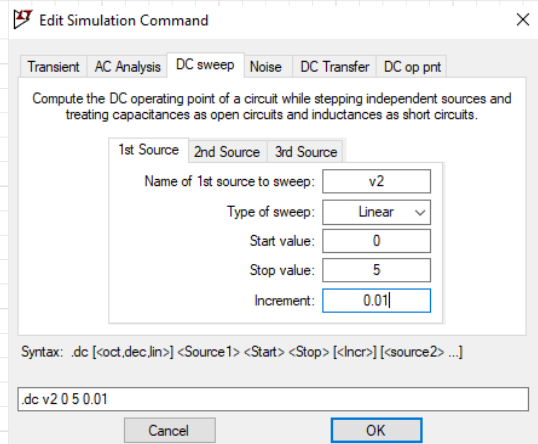
You will need to go to the component section of LTSpice and choose: Voltage (Source), nmos (transistor). You can get resistors from the main menu bar (Windows version).

Objective: Determine the values for resistors R1, R2 and voltage V1. Simulate and show your demonstrator once you have achieved this.

Steps: How to achieve this:

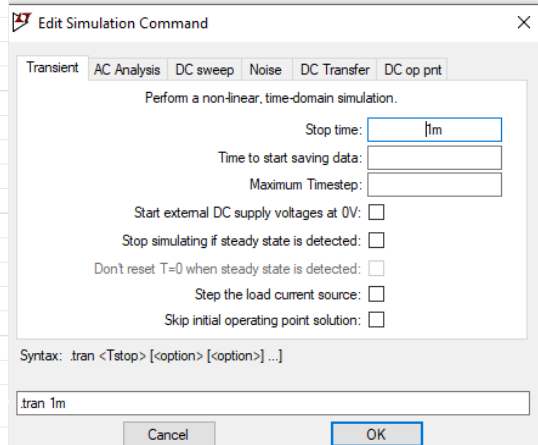
- (1) Connect voltage V1 to the drain of the transistor to provide a V_{ds} . Then connect V2 to the gate to provide a V_{gs} .

Then to a DC simulation – scanning V2 between 0 and 5V (with V1 = 5V). From this graph you can work out which voltage V_{gs} is required to get a drain current $I_d = 40\mu\text{A}$.



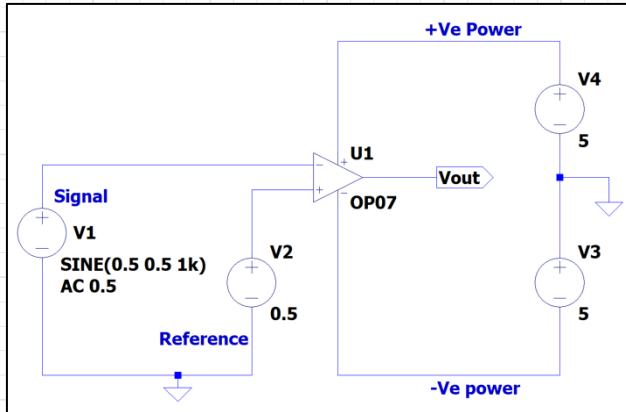
Steps: How to achieve this:

- (2) Reconfigure the circuit with resistors, and use appropriate values to get the required voltage. Use a transient simulation (1ms) to double check that your V_{gs} is correct and that you are getting $I_d = 40\mu\text{A}$



Lab 2: Amplifiers

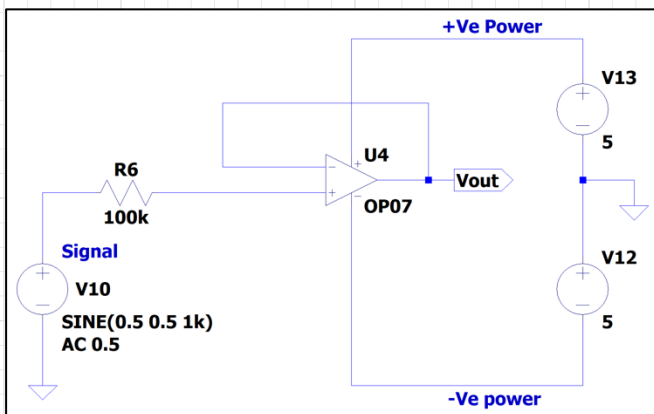
Task 8: An Op Amp as a comparator



The first task is to explore the operational amplifier as a comparator. Use a OP07 OpAmp model. Set: Bias voltages V3, V4 = 5V
Reference voltage V2 = 0.5V
Signal voltage V1 = 0.5V AC + 0.5V offset @ 1kHz

1. Do a transient simulation for 10ms. Overlay the output Voltage V_{OUT} and signal Voltage V1. Explain the result.

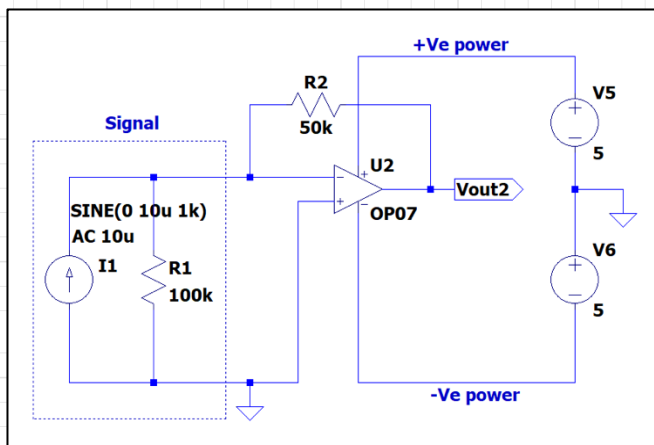
Task 9: Voltage buffer



Now lets explore the voltage buffer. Use a similar set up as before but connect the negative terminal directly to the output. For the input signal, set: V1 = 0.5V AC + 0.5V offset @ 1kHz
Series resistance R6 = 100 kΩ

1. Compare the output voltage with the input voltage.
2. What happens if you change the amplitude of the signal to greater than 5V?

Task 10: Transimpedance amplifier

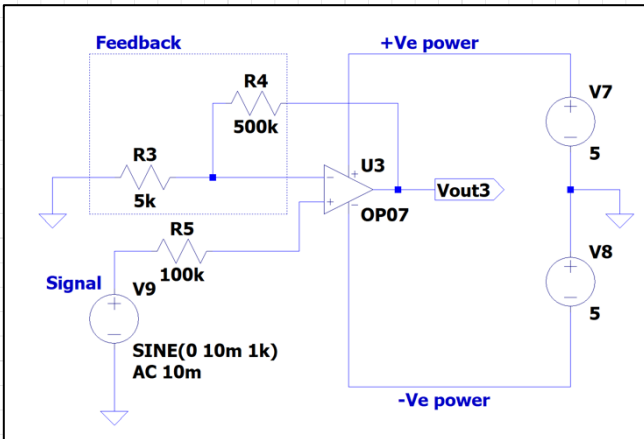


Now let's consider using an operational amplifier to create a transimpedance amplifier. This type of amplifier can be used to convert current into a voltage. E.g. photodiode current in a camera to a voltage level suitable to provide to an analog to digital converter. Set:
Bias voltages V5, V6 = 5 V
Signal current I1 = 10 μA, 1kHz AC,
shunt resistance R1 = 100 kΩ
Feedback resistance R2 = 50 kΩ

Objective: Do a time domain transient analysis to plot the output voltage waveform with the input current waveform. Then determine the gain in V/A!

Lab 2: Amplifiers

Task 11 Non-inverting amplifier



Now let's look at a non-inverting voltage amplifier.

Bias voltages $V_3, V_4 = 5\text{ V}$

Set Signal voltage $V_9 = 10\text{mV (AC)}$, 1kHz

Signal series resistance $R_5 = 100\text{ k}\Omega$

Feedback resistors $R_3 = 5\text{ k}\Omega$, $R_4 = 500\text{ k}\Omega$

1. Perform a transient simulation and explore how the output voltage varies with the signal when R_4 is equal to $50\text{ k}\Omega$ to $500\text{ k}\Omega$
2. Do an AC simulation and obtain a bode plot (gain vs frequency). Explain the result.

Circuit Lab: Reflections

Some additional things to ponder:

Task 1:

1. Imagine you are trying to record a small voltage from a heart through an electrode. What happens if the electrode impedance is higher than the input impedance of your recording circuit (load)?

Task 2:

2. If you want to record a signal with a frequency range of 1-20 Hz, how do you remove high frequency noise?

Task 3:

3. If you want to separate a low frequency component from a signal, what would you do?

Task 4:

4. How do you remove the negative component of a signal or power supply?

Task 5:

5. How do you add a DC offset to an AC signal? (note, this is important later on!)

Task 6:

6. Although bipolar junction transistors (BJTs) are found in many discrete amplifiers, the microelectronics industry is now almost entirely using MOS Field effect transistors (MOSFETS).

Task 7:

1. Imagine you need to threshold a signal. Anything above a set value should trigger an action. What circuit would you use?

Task 8:

2. Analog to digital converters typically found on microcontroller units require voltage inputs. How would you convert a small current into a voltage to input to an ADC? Why is this better than simply putting the current across a large resistor?

Task 9:

3. Digital to analog converters will create a voltage, but typically cannot drive a large current. So how should you connect your DAC to a small load (which will have a large current)?

Task 10:

4. Many sensors will give voltages in the mV range or even smaller. However, a 10 bit ADC with a 2V dynamic range would represent each bit in 2mV steps. i.e. waveforms significantly smaller than 2V will not digitize well. So how do you increase your signal?